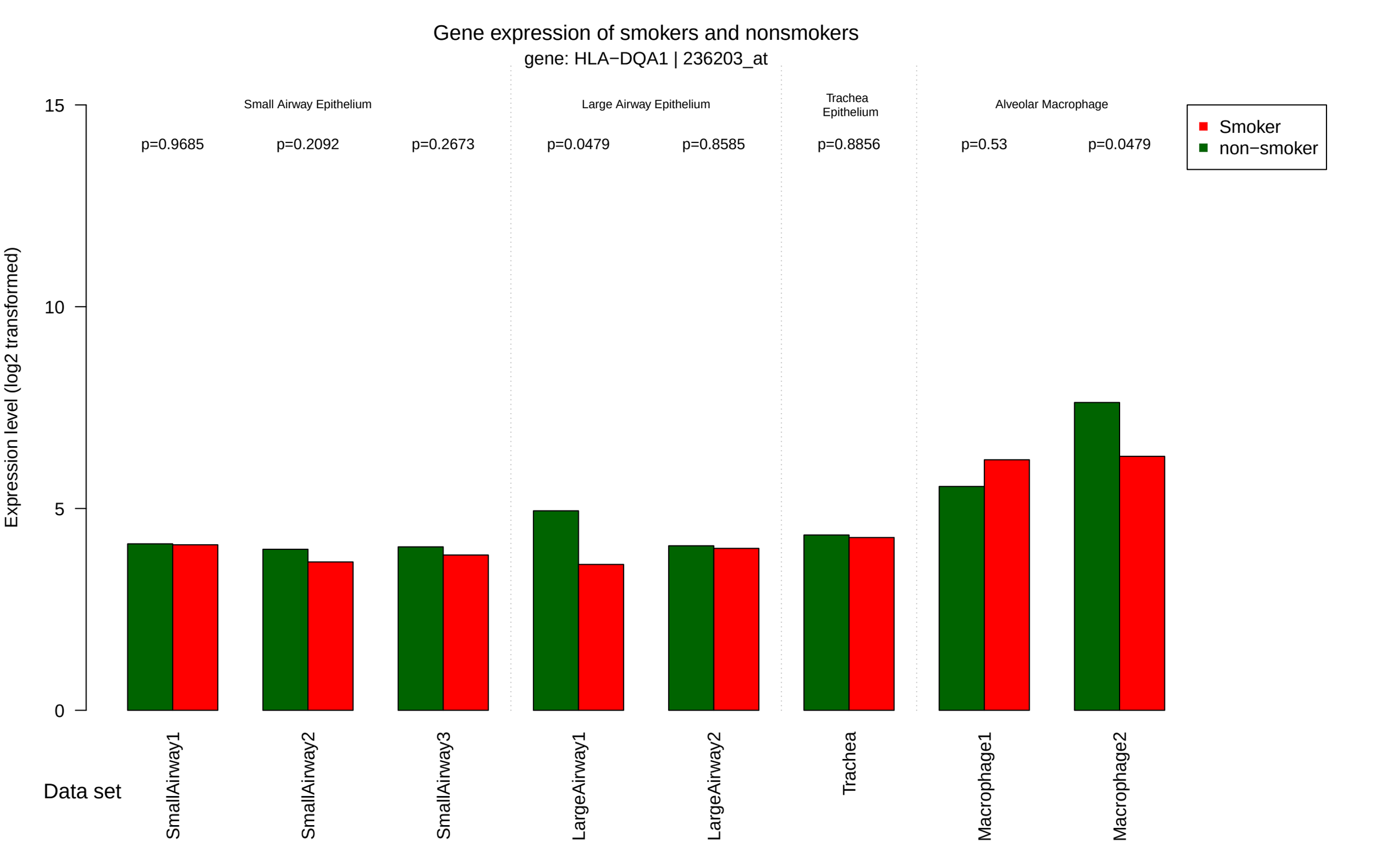
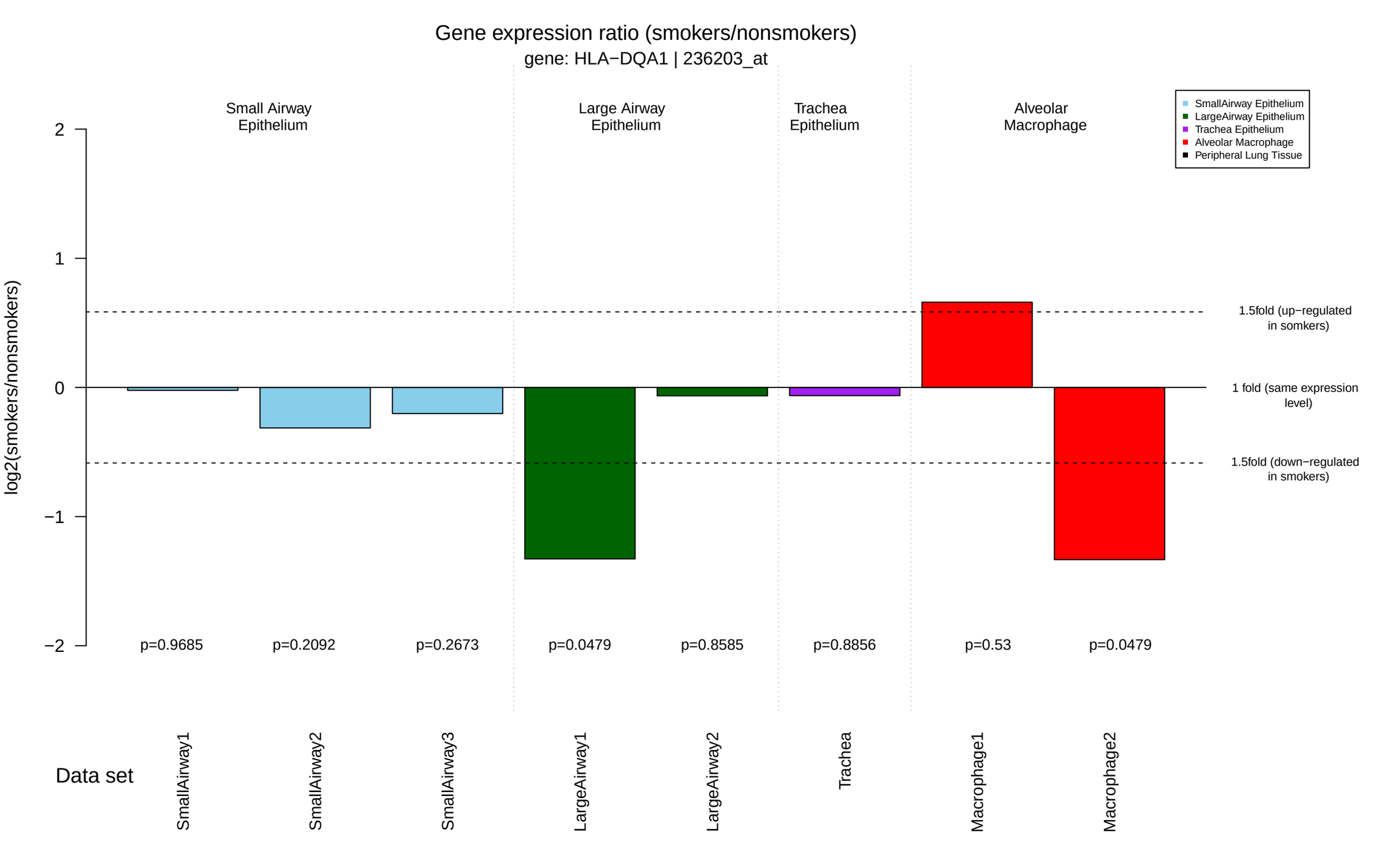
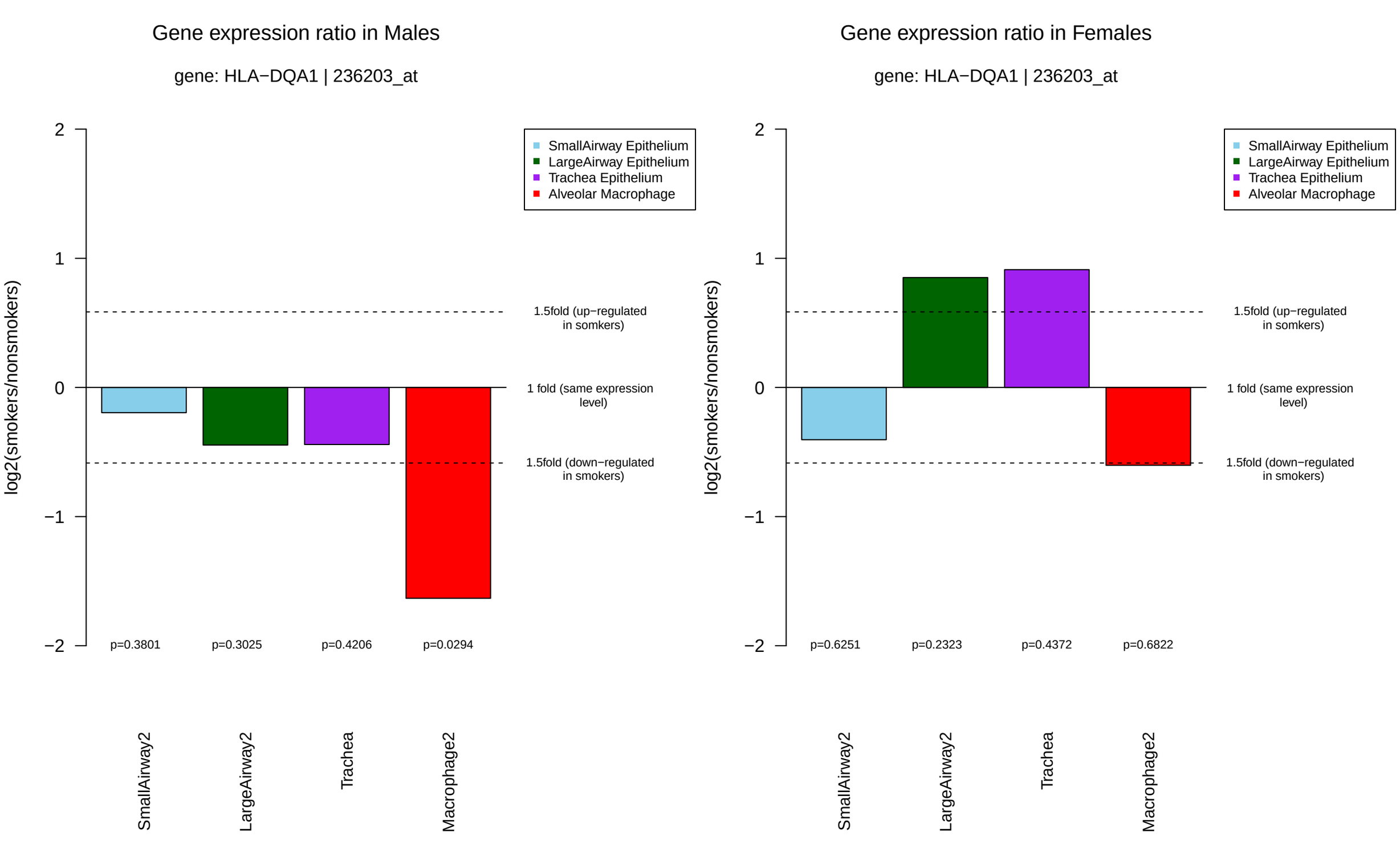
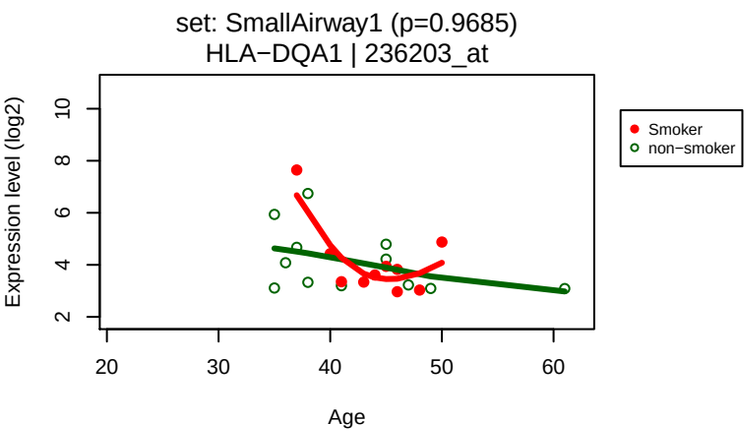
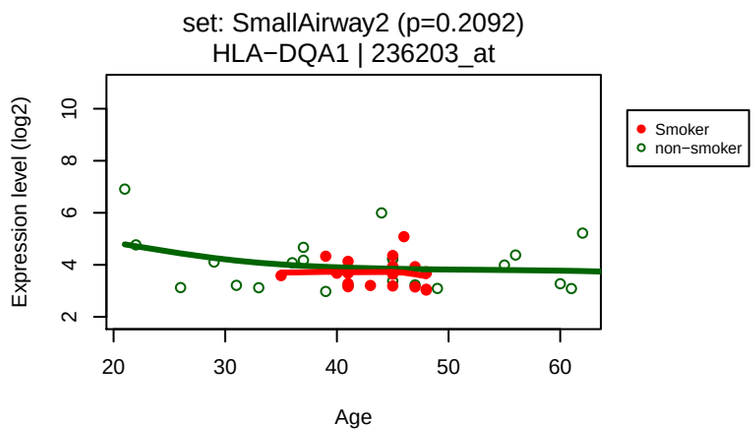
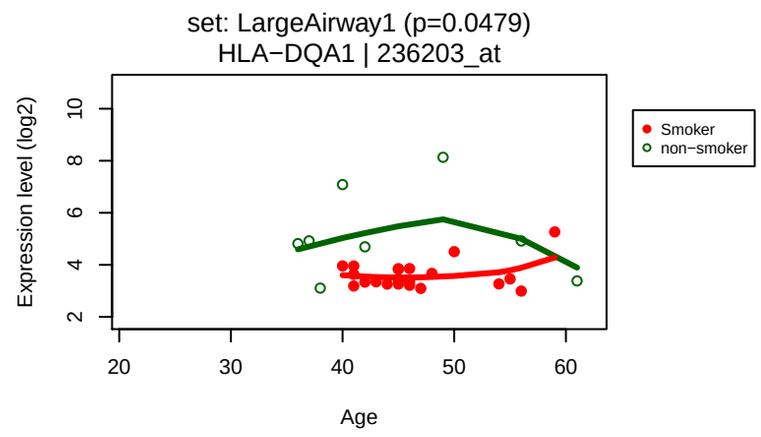

Gene expression of smokers and nonsmokers
gene: HLA−DQA1 | 236203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9685
p=0.2092
p=0.2673
p=0.0479
p=0.8585
p=0.8856
p=0.53
p=0.0479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

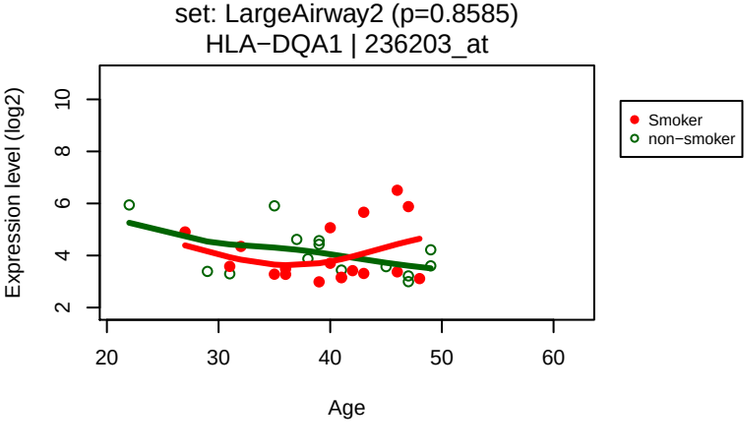

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQA1 | 236203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9685
p=0.2092
p=0.2673
p=0.0479
p=0.8585
p=0.8856
p=0.53
p=0.0479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

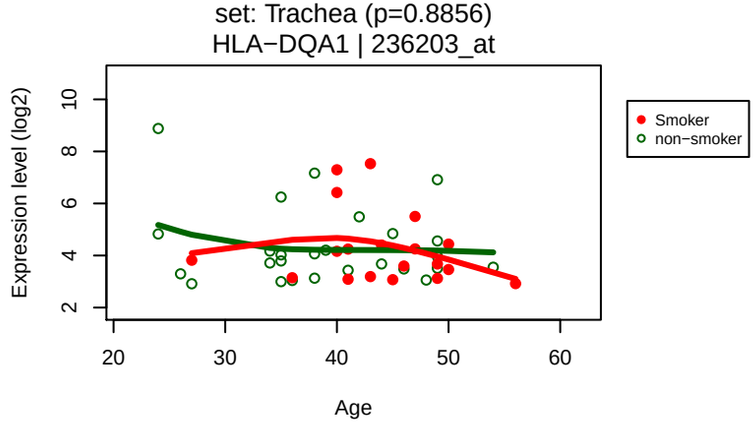

Gene expression ratio in Males
gene: HLA−DQA1 | 236203_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3801
p=0.3025
p=0.4206
p=0.0294
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQA1 | 236203_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6251
p=0.2323
p=0.4372
p=0.6822
SmallAirway2
LargeAirway2
Trachea
Macrophage2

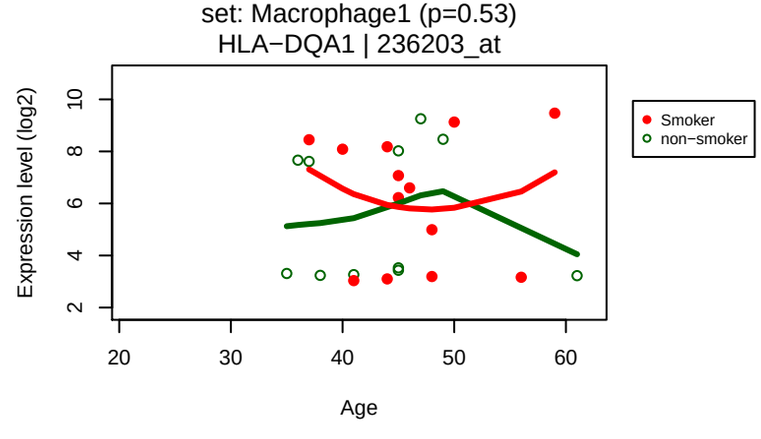

set: SmallAirway1 (p=0.9685)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker

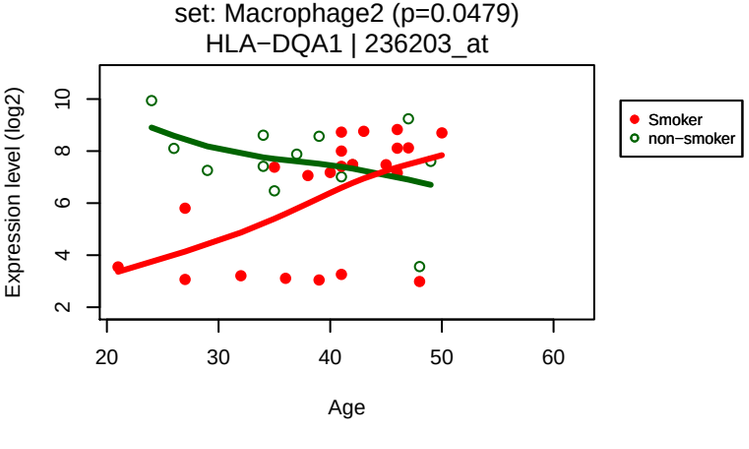

set: SmallAirway2 (p=0.2092)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker

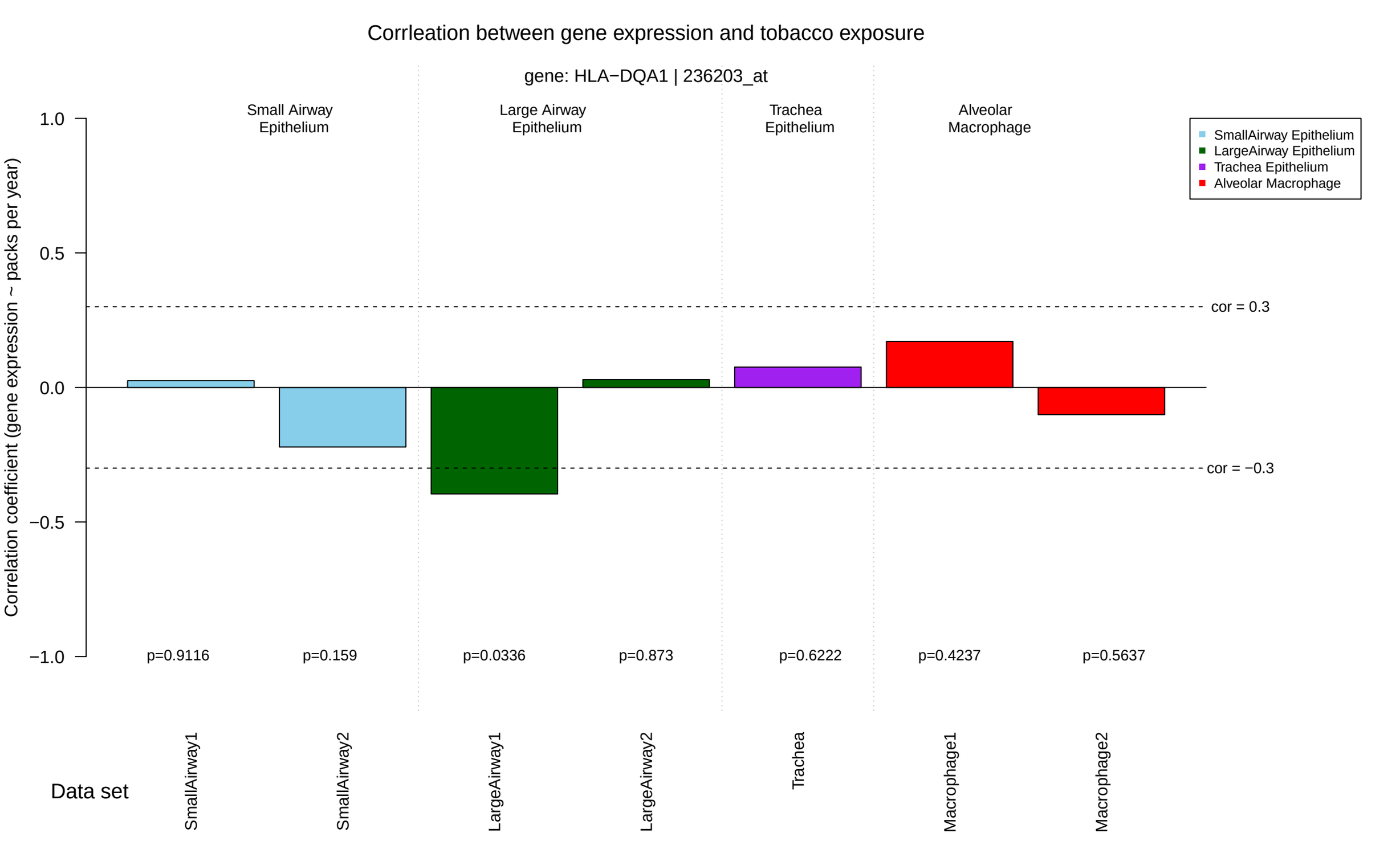

set: LargeAirway1 (p=0.0479)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8585)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8856)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.53)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0479)
HLA−DQA1 | 236203_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQA1 | 236203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9116
p=0.159
p=0.0336
p=0.873
p=0.6222
p=0.4237
p=0.5637
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2